Developing Airmen and Guardians With Games for Enhanced Readiness (DAGGER): Using Minecraft to Build Cadet Teamwork and Communication Skills

Abstract

Although many military training and education programs can leverage physical obstacle courses and higher-fidelity simulations to practice for team-based missions, Air Force Reserve Officer Training Course (AFROTC) programs have historically been limited by physical space and resource limitations. In this context, team-based activities in AFROTC Leadership Lab (LLAB, AFROTC's required leadership course) are often limited to discussion of hypothetical scenarios, potentially limiting opportunities for cadets to practice teamwork and interpersonal skills. In FY2024-2025 AFROTC evaluated a pilot program (Developing Airmen and Guardians with Games for Enhanced Readiness) in which select AFROTC detachments participated in team activities in Minecraft (collaborative building, navigation, and escape room missions) and used a web-based competency rating tool to provide cadets feedback on competencies demonstrated during the Minecraft activities. Cadet comments, as recorded in the web-based rating tool, highlighted the importance of team communication, planning, and delegation to perform the activities effectively, and survey results showed over 90% of participating cadets perceived the Minecraft activities as useful to practice Communication, Teamwork, and Decision-Making. Additionally, all cadets who used the Minecraft activities in combination with the web-based rating tool indicated they received useful feedback on these competencies. Cadets who used the web-based rating tool to give and receive peer feedback indicated that doing so helped them reflect on the competencies and their own strengths and weaknesses. Notably, cadets who had never, or rarely, played Minecraft or other online games before generally perceived the

DAGGER Minecraft activities as positively as those with more extensive Minecraft or gaming experience.

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Across military and civilian settings, organizational leaders increasingly recognize that it is not sufficient for personnel to be trained on technical knowledge and technical skills; rather, personnel must also develop "soft skills" – also termed 21st Century Skills – to be successful in the workforce and more fully contribute in military roles (see for example, Herman et al., 2019; Robson et al., 2024). 21st century skills are typically conceptualized as including both cognitive skills, such as critical thinking, problem-solving, creativity, and information literacy, and interpersonal skills such as adaptability, collaboration, leadership, and communication (National Research Council, 2012). Within the military, such soft skills have been referred to by various terms, to include Foundational Competencies (U.S. Air Force) and Talent Attributes (U.S. Army) and specific soft skills (Foundational Competencies) have been identified and validated through large-scale studies as widely important for success across a military career, across ranks and functional areas (see Barron & Rolwes, 2020; Barelka et al., 2019; Royston et al., 2022). Although such skills are teachable (National Research Council, 2012; Robson et al., 2024), developing 21st century skills requires a different kind of teaching than has historically been emphasized in the military (Herman et al., 2019). Scholars have emphasized the importance of shifting from traditional modes of instruction, such as lectures or readings, that may help a learner understand a solution to a specific problem, to experiential project and game-based learning to promote the development of interpersonal skills and transfer of learning to new situations (see Herman et al., 2019).

The U.S. military's Reserve Officer Training Corps (ROTC) programs enable students (cadets) to train to become military officers while attending a civilian college or university.

These programs provide leadership education and typically allow cadets to learn about the military, from military officers, prior to committing to a military career. Although Air Force ROTC cadets are expected to develop some military-specific knowledge during the program, they are also expected to practice, receive feedback, and develop on soft skills ("Foundational Competencies") valued by the military such as Teamwork and Communication.

Historically, Air Force ROTC's leadership education course ("Leadership Lab") has relied heavily on group discussions of hypothetical scenarios (such as those involving ethical dilemmas, or prioritization of conflicting needs of team members) to educate cadets on competencies such as Teamwork and Communication. Recognizing a potential need for more active and experiential learning that could be accomplished with limited physical space and resource constraints, Air Force ROTC piloted a program to evaluate how collaborative videogame play could potentially support soft skill development. Specifically, in this study we evaluate whether a popular, low-fidelity commercial gaming platform (Minecraft for Education) could be effectively leveraged in brief team activities to help cadets practice and receive useful feedback on widely generalizable soft skills in the context of a leadership education course.

Minecraft for Education

Minecraft is the best-selling videogame of all-time, having sold over 350 million copies as of 2025 (Guiness Word Records, 2025). Minecraft is a versatile sandbox game that includes building and survival elements, and can be used in a number of ways to include building one's own virtual world, and engaging in goal-directed "quests" as in role-playing games, including those requiring multiple players to accomplish a team goal. Although the graphics are not realistic, Minecraft allows for navigation across obstacles within a 3-dimensional (albeit virtual) space providing a potential online alternative to the types of in-person team exercises that are

often conducted on physical obstacle courses in Professional Military Education ("Project X") to simulate military team operations (e.g., gaps to cross, minefields to navigate) (see U.S. Air Force Academy, 1996).

In 2016, Microsoft released a "Minecraft: Education Edition," specifically targeted at teachers (Dwilton, 2020). Although primarily used to teach students in specific K-12 content areas (e.g., math, computer science, social studies), Microsoft's marketing materials have emphasized the development of 21st century skills, with claims that that Minecraft can help students build creativity and problem solving skills, as well as "socio-emotional skills" such as empathy (Microsoft, 2026). A recent systematic research review found "promising evidence that digital game-based learning with Minecraft can improve cognitive (i.e., creativity, spatial reasoning) and academic outcomes (i.e., mathematics and science) in children, adolescents, and young adults" (Slattery, 2025, p. 26), but noted few studies had evaluated the use of Minecraft for building interpersonal skills like collaboration.

Nevertheless, team-based exercises in Minecraft are increasingly used in K-12 and university education (see Di Leo & Traetta, 2025; Lane & Yi, 2017), and may have the potential to support interpersonal skill development (see, for example, Hebert & Jenson, 2020; Wendel et al., 2013). We could locate only one study that has evaluated the use of Minecraft-based team activities for interpersonal or collaborative skills in a university context (de Weck et al., 2022). In the study, Massachusetts Institute of Technology (MIT) undergraduates, working in virtual teams of 5-6 students, participated in a highly structured Minecraft building activity over two 60-minute sessions then completed an end-of-activity survey. Survey responses indicated that, after participating in the team activities, 75-80% of students felt more confident in their ability to "describe the importance of clear communication and decision making during projects" and

"explain the importance of dividing up large tasks and assigning them based on team member strengths" (de Weck et al., 2022, p. 18).

Though the results were promising in demonstrating that team-based Minecraft building activities could help college students appreciate the importance of communication, decision-making, and (certain aspects of) teamwork, the MIT intervention did not aim to provide students with feedback on their own skills in these areas (de Weck et al., 2022). Hence it is unclear the extent to which the Minecraft team activities provided a useful basis for giving and receiving feedback on these competencies, or actually improving students' skills in these areas. Additionally, although the authors acknowledged concerns about integrating students without prior Minecraft experience (de Weck et al. 2022), they did not assess student perceptions separately based on prior gaming experience. Further, the de Weck et al. (2022) study was conducted with students at a very selective, engineering-focused university, in which students may have been more motivated to engage in building activities than would typical students.

Potential Use of Virtual Game Play to Substitute for In-Person Physical Obstacle Course Exercises in Military Education

In-person physical obstacle courses have been used since the 1950s within Air Force Professional Military Education as a means to assess and develop leadership skills (Project X Leadership Reaction Course; see U.S. Air Force Academy, 1996; Schellhous, 1998). Originally developed for England's Royal Military Academy as part of their selection process, the exercises were later adapted and modified by U.S. Air Force Squadron Officer School faculty, with the criteria that the challenges presented be realistic and require a small team (minimum of four members) to work together to find a solution (Schellhous, 1998). In addition to their use in Squadron Officer School, such exercises have been used in educational programs for U.S. Air

Force Academy cadets, and newly accessed officers in Officer Training School; similar exercises have also been implemented by the U.S. Army and U.S. Marine Corps, and have been used within the German military since at least the 1930s (Schellhous, 1998). Normally, as a required experiential component of their education, in-person physical obstacle courses are used, and military students are tasked to work in small teams to address a series of simulated military scenarios requiring physically climbing, balancing, lifting, jumping, and otherwise safely navigating physical obstacles to enable all members of their team to meet a shared goal. Although the exercises have a physical component, the primary intent is not to test members' physical fitness, but rather to challenge students to engage in problem-solving as a team (Schellhous, 1998).

Through guided post-exercise instructor-led group debriefs, students are directly prompted to reflect on soft skills, such as teamwork, communication, and decision-making demonstrated through the exercises (U.S. Air Force Academy, 1996). For example, instructors may ask students how they delegated team roles in the exercise, how they decided the approach they would take, and how they addressed conflicts between team members (U.S. Air Force Academy, 1996). As originally conceptualized within the U.S. Air Force, the exercises were intended, at least in part, to "identify the leader strengths and weaknesses of each student as a basis for self-evaluation and consequent self-improvement" (Schellhous, 1998, p. 11). However, although students may provide verbal feedback to their teammates during the de-brief, normally students are not expected to directly rate themselves, or their teammates, on either their individual performance, or the soft skills they may have (effectively or ineffectively) demonstrated in the exercise. Research has shown that, overall, teams that attempt the exercises at the beginning of Squadron Officer School are less likely to solve the problems presented than

teams that attempt the exercises at the end of the course (Schellhous, 1998). However, we could locate no studies that sought feedback from participating students, or that actually evaluated the extent to which students view these types of exercises as allowing them to effectively practice specific competencies or as a means to receive meaningful feedback in these areas.

One small qualitative case study based on 18 military students demonstrated that the types of challenges normally presented in the in-person physical obstacle course in Squadron Officer School (Project X; requiring teams to cross a gap and navigate physical obstacles in a simulated minefield) could be effectively re-created in a low-fidelity virtual environment in which students appeared as avatars and used standard computer inputs to navigate within a virtual space (Clayton, 2017). That is, it was not necessary to physically carry or physically lift a teammate to cross an obstacle in order to create an environment requiring application of soft skills such as teamwork, communication, and decision-making. Rather, as students navigated a virtual world in real-time, their verbal interactions with other players demonstrated elements of teamwork and leadership such as setting expectations, directing teammates, and providing encouragement (Clayton, 2017). Although students, when prompted by the investigator in interviews, could provide examples of soft skills applied within the virtual environment (Clayton, 2017), it is unclear the extent to which their interactions with teammates during the exercise actually provided a sufficient basis for students to notice and meaningfully evaluate the soft skills of individual teammates.

Interest in Gaming Among Military Populations

Many young people, including those who enter the military, play videogames regularly at home, and may be motivated to use videogames for training purposes (Prensky, 2001). Trainee motivation, in turn, has shown to increase time on task in training, and subsequent learning

outcomes (Colquitt et al., 2000; Mathieu, et al., 1992). Although videogames can be motivating for many, the proportion of military members who enjoy videogames may be lower than is conventionally believed. Previous research in military cadet samples has shown that prior videogame experience (frequency of videogame use) is positively related to satisfaction with game-based training, ease of use in navigating training game interfaces, and training game performance (Orvis et al., 2007, 2009). These relationships were stronger when trainees had prior experience with videogames that shared more similar features to the training game (such as pace, interface, perspective) (Orvis et al., 2007). Prior experience with videogames focused on puzzles and card games, for example, was unrelated to training outcomes with a first-person shooter game (Orvis et al., 2007). Hence an important consideration in evaluating the DAGGER program was whether the program would be perceived positively even by cadets who lacked prior experience with Minecraft and other team-based online games.

Research Questions

In the current study we sought to evaluate two primary research questions, based on (i) an audit of peer feedback collected using the web-based rating tool, and (ii) a survey of cadets from AFROTC detachments that had used the Minecraft activities (to include those who had participated in the activities without use of the web-based rating tool):

(1) To what extent would Minecraft-based team activities provide a useful opportunity for military cadets to practice and develop on soft skills such as Teamwork and Communication?

(2) To what extent would Minecraft-based team activities provide a useful basis for self-assessment and peer feedback on these soft skills?

We also considered the following secondary questions:

(A) To what extent do current military cadets have prior experience with Minecraft and other types of online games?

(B) To what extent do cadets with little or no Minecraft or applicable gaming experience perceive team-based Minecraft activities as useful for competency development?

Method

Overview of DAGGER Implementation in AFROTC Leadership Lab

The DAGGER initiative was implemented at select AFROTC detachments as part of Leadership Lab, a credit-awarding course offered at universities with an AFROTC program, typically offered each semester and graded pass/fail. Attendance is mandatory for enrolled students. HQ AFROTC managed the pilot program, and detachment commanders interested in participating opted-in on a first-come, first-serve basis. Instructions for implementing collaborative Minecraft team activities in LLAB (Building, Navigation, and Online Escape Room activities) and using a web-based tool for peer feedback were made available on the DAGGER website. Detachment commanders and LLAB instructors could decide which Minecraft activities they wanted to implement, and the extent to which they wanted to integrate the web-based rating tool into LLAB (if at all).

Although individual AFROTC programs had discretion as to which team activities were offered in Leadership Lab, all shared a common requirement to ensure enrolled cadets met specified learning outcomes to meet Air Force standards (see AFROTCI 36-2011 v1). These standards included:

- Recall the Airmen's Foundational Competencies (8.5)
- Demonstrate effective followership (8.1) and team leadership (8.2)
- Demonstrate the principles of an effective debrief (8.4)
- Practice leadership evaluation (8.9)

The Minecraft LLAB exercises were intended to provide an opportunity for participants to demonstrate followership and team leadership, and engage in the process of evaluating others (teammates) and providing constructive feedback (principles of an effective debrief). They were also intended to help cadets reflect on (and recall) defined Foundational Competencies such as Teamwork, Communication, and Flexibility (see Appendix 1 and 2)

Minecraft Activity Procedure

Cadets typically completed one exercise per week while enrolled in AFROTC Leadership Lab, which could be either an activity in Minecraft or a legacy LLAB activity (typically involving discussion of a written scenario). Each exercise was designed to be completed within 20-30 minutes, to allow time for competency ratings and a group debrief. Exercises were completed in small groups (teams), with AFROTC guidance recommending three or four participants per team, but allowing for flexibility. Individual AFROTC programs selected which Minecraft and/or legacy LLAB activities to include in their university's course from a list of options. Three types of Minecraft exercises were offered as part of the DAGGER program: building, building navigation, and escape room activities. See sample descriptions in the Appendix 3 and 4.

As shown in Figure 1, Minecraft was accessed individually by each cadet on an individual laptop (not shared with other cadets). However, cadets could interact both within and

outside of the virtual game environment. Although classroom layout varied somewhat by university, cadets were physically co-located within a classroom space where they could see their teammates and communicate directly with one another in-person outside of the virtual Minecraft environment.

Web-Based Competency Rating Procedure

Detachments had the option of using a web-based rating tool for cadets to rate themselves on the Foundational Competencies and provide peer feedback (see competency definitions and example rating criteria in Appendix). In detachments that used the web-based tool, cadets rated themselves and up to three peers on three of eight competencies based on the Leadership Lab team activities, with the specific competencies rated in an individual exercise identified a priori and communicated to cadets by instructors before they began the exercise (see Appendix 1 and 2 with the competencies rated for each exercise). The web-based rating tool prompted each cadet to rate themselves and up to three peers (teammates) immediately following each exercise (up to 12 total ratings per exercise). Cadets first selected which cadet they were rating on from the pre-populated list of teammates that participated in the exercise, and then provided ratings and comments in succession for each of the three competencies. Competency ratings were made on a 1 to 5 scale, and then cadets were prompted to provide narrative feedback to justify each rating (see screenshots inFigure 2). Cadets were also prompted to provide an overall rating of their team's performance in the activity with accompanying comments. The process of providing inputs was designed to take no more than five minutes.

Immediately after cadets submitted their ratings, the app displayed a "Justification Score" for each competency. This score was not based on how they were rated by others, or the numeric ratings they provided, but instead was based on the narrative comments they provided for each

competency. “Justification Scores” were computed based on the length of the narrative comments they provided, and the uniqueness of each comment-- such that providing the same comment for each of the individuals they rated resulted in a lower Justification Scores. The display of the Justification Scores was intended to encourage cadets to more deeply evaluate cadets and provide higher quality feedback when completing ratings on exercises later in the semester.

Next, the app enabled cadets to view their self-rating on each competency juxtaposed with how their peers rated them (average peer rating for each competency). To prevent cadets from reading any narrative comments about themselves that could potentially be hurtful, the narrative comments were only viewable by the course instructors (who conducted the group debrief) rather than by the individual cadets who were rated. Course instructors reviewed narrative comments to help guide the group debrief (“hot wash”) after the team activities and, in some cases, met privately with some cadets to discuss the feedback.

Audit of Ratings and Peer Feedback Collected Using the Web-Based Rating Tool. We obtained de-identified access to all self-, peer-, and team ratings and feedback that cadets had provided in the web-based rating tool at the largest participating detachment (Detachment 150, University of Florida). After an initial Minecraft orientation activity, the detachment had used the web-based rating tool in combination with three Minecraft building navigation activities and four Minecraft escape room activities over the course of a semester (March – June 2024),. Using the tool over successive LLAB sessions, a total of 15 cadets from the detachment (organized in teams of three or four) provided multiple ratings of themselves and their teammates on nine competencies based on performance in the Minecraft activities. Self- and peer- competency ratings (which were recorded immediately after each Minecraft activity) were each accompanied by narrative

feedback, and ratings and comments on team performance. We analyzed word frequency in the cadet comments on team performance in each of the two types of Minecraft exercises, and reviewed the peer comments corresponding to ratings of higher vs. lower ratings of team performance, and higher vs. lower ratings of peers on each of the competencies. *Survey Administration*. At the conclusion of the DAGGER program trial, we also sought to obtain direct feedback from cadets on the perceived utility of the Minecraft activities and the web-based rating tool, and to distinguish reactions based on cadets' prior Minecraft and gaming experience. In September 2025, HQ AFROTC identified the email addresses of cadets who had created an account for the DAGGER web-based rating tool in FY2024 or FY2025 at one of the participating AFROTC detachments. HQ AFROTC also emailed a QR code to the 36 detachment commanders who had opted into the DAGGER program that could be posted in classrooms to encourage cadet participation in the anonymous survey.

Survey Participants. Overall we received 149 usable student survey responses, including responses from 145 current AFROTC cadets, and five other DAF members (i.e., former cadets who had since commissioned) (see Table 1). Students who initiated the survey but were no longer affiliated with AFROTC or DAF were ineligible and were exited out of the survey. As shown in Table 1, the sample included predominantly cadets in their first or second year of the AFROTC program (AS100 and AS200, $N = 92$); these are typically freshmen and sophomores not yet committed to enter the Department of Air Force upon graduation. In contrast, AS 300/400/700/800/900 cadets are upperclassmen who have made a service commitment and been selected into the Professional Officer Course.

Although all respondents completed background survey questions on recreational online gaming experience, a much smaller number of respondents indicated that they had participated in

the Minecraft activities during LLAB (N = 35) and/or used the web-based rating tool (N = 19). Because of the survey timing (administration of the survey after the DAGGER trial concluded), many of the AFROTC cadets who accessed the survey QR code had not been enrolled in LLAB at the time that the DAGGER activities were offered at their detachment. Though we analyzed responses to the recreational online gaming experience section of the survey from all respondents, we only analyzed survey responses regarding the LLAB Minecraft activities and web-based rating tool from cadets who had participated in the Minecraft activities and/or made ratings based on LLAB activities during LLAB sessions. Cadets who observed others performing the Minecraft activities-- but did not participate themselves—and those who used the web-based tool to make peer ratings as homework after LLAB rather than during the LLAB class session itself were excluded from completion of the LLAB survey questions. Among those who had directly participated in Minecraft activities during LLAB, 40% participated solely in Minecraft building activities, 20% solely in Minecraft escape room activities, 14.29% solely in Minecraft navigation activities, and 25.71% participated in multiple Minecraft activity types (e.g., both building and navigation activities).

Survey Measures

Prior Videogame Experience. All respondents completed questions on their prior experience with Minecraft and other online games (prior to participation in DAGGER). They were asked the frequency with which they had previously used (i) Minecraft, (ii) Online single-player games, (iii) Online team-based games, (iv) Online building games, and (v) Online escape room games. Frequency was measured on a six-point scale ranging from 0 = Never to 5 = Very Frequently.

LLAB Minecraft Activities. Cadets who participated in Minecraft activities in LLAB were separately asked the extent to which the Minecraft activities in LLAB allowed them to (i) practice and (ii) receive useful feedback on each of eight Foundational Competencies: (a) Accountability, (b) Decision Making, (c) Developing Others, (d) Communication, (e) Flexibility, (f) Self-Control, (g) Teamwork, and (h) Results Focus. Students could also provide optional narrative comments. The survey included the following prompts: (i) "Please provide any comments about your experiences with the Minecraft activities in LLAB," and (ii) "Please provide any additional comments about the DAGGER program."

Use of the Web-Based Rating Tool. Cadets who used the web-based rating tool during LLAB rated the extent to which they agreed or disagreed with three items regarding providing self- and peer-ratings: (i) "The process of rating myself in LLAB helped me reflect on the Foundational Competencies" (ii) "The process of rating my teammates in LLAB helped me reflect on the Foundational Competencies and (iii) "Having to provide comments to justify my ratings of teammates on each competency helped me reflect on the Foundational Competencies." Cadets also rated one item regarding receiving peer feedback: "Viewing how my teammates rated me on each competency in LLAB helped me better understand my competency strengths and weaknesses."

Results

Team Performance in the Minecraft Activities (Web-Based Rating Tool). Immediately after each Minecraft team activity, participating cadets at Detachment 150 had rated and provided comments on how their team (3 to 4 cadets) had performed in the activity using the web-based rating tool. Team performance ratings (1 to 5) averaged 3.83 (SD = 0.67) in the three building navigation activities, and 3.90 (SD = 0.75) in the four escape room activities. Narrative

comments on team performance in each of the three Minecraft building navigation activities ranged from 6 to 82 words (median 25) per cadet; and 8 to 76 words (median 23) in each of the four Minecraft escape room activities. Collectively, the detachment had provided 25comments regarding team performance in the building navigation activities (887 total words) and 31 comments (934 total words) regarding team performance in the escape room activities (separate from feedback evaluating the competencies of individual teammates that is described in the next section).

We began by analyzing word frequency in cadet comments on team performance to identify overall themes. As shown in Figure 3, “communication” (and its direct variants, “communicate,” “communicated,” and “communicates”) was among the most commonly occurring words that cadets used when describing team performance for both the Minecraft building navigation and escape room activities; other commonly occurring word groups included teamwork, completion, mission, and objectives. Comments on team performance differed somewhat by Minecraft activity type; escape room comments more frequently referenced usage (i.e., of resources or helpers) and quickness, while building navigation activity comments more frequently referenced planning. See Figure 3.

We next sought to distinguish factors contributing to successful performance of the Minecraft activities by contrasting performance comments from cadet teams with the highest vs. lowest average ratings of team performance (i.e., as rated by members of the team). Comments on team performance in the Minecraft building navigation and escape room activities from members of the higher and lower performing teams, respectively, appear in Table 2. As shown in Table 2, members of the highest-performing team in the Minecraft Navigation activities attributed their success in the activities to strong teamwork and leadership, establishing goals and

role delegation, clear communication, and working together to find creative solutions. Members of the lower-performing team, that eventually completed the activity after multiple tries, indicated they could have done better communicating as a team to think critically and distinguish information that was and was not relevant to solving the problem. Members of the higher-performing team in the Minecraft Navigation activities described communicating to effectively delegate and distribute work. By contrast, members of the lower-performing team described initially failing to clarify their objectives and establish a plan before beginning execution.

Feedback on Peer Competencies Based on the Minecraft Activities (Web-Based Rating Tool). Instructions within the web-based competency rating tool prompted cadets to justify their numeric (1-5) ratings of each teammate on each of three target competencies demonstrated during each Minecraft exercise with narrative comments. Narrative comments justifying ratings of each teammate on each competency ranged from 7 to 78 words (median 18) per cadet evaluated. Overall, as shown in Tables 3 and 4, cadets generally provided substantive feedback that could be meaningfully distinguished based on numeric ratings of Accountability, Communication, Self-Control, Results Focus, Teamwork, Flexibility, and Decision Making. However, some cadets commented that it was not possible to meaningfully observe or practice Developing Others or Fostering Inclusion within the constraints of the escape room missions. Consistent with this, comments on Communication, Decision Making, Flexibility, and Teamwork generally included more unique words (i.e., not repeating the same comment for more than one peer, and suggesting greater differentiation in feedback) than comments on Develops Others, Results Focus, Fosters Inclusion, and Self-Control. See Table 5.

Cadet Survey: Prior Gaming Experience. As shown in Table 6, over 75% of cadets had played Minecraft outside of the DAGGER program. However, the extent of cadets' experience

with Minecraft varied considerably; only 24.83% of cadets indicated they played Minecraft "Frequently" or "Very Frequently." The proportion of cadets who played any online building games (including but not limited to Minecraft) was similar (23.61% of cadets never played online building games, while 28.47% of cadets did so frequently or very frequently). Online escape room games were played substantially less frequently than building games (44.14% had never played online escape room games, and only 7.59% had done so frequently or very frequently). Overall 45.52% of cadets indicated they played team-based online games "Frequently" or "Very Frequently" (exceeding the 42.76% of cadets who frequently or very frequently played single-player online games).

Cadet Survey: Perceptions of LLAB Minecraft Activities. As shown in Table 7, most cadets viewed the LLAB Minecraft team activities as useful to practice and receive feedback on the Foundational Competencies (see competency definitions and rubrics as presented in LLAB in Appendix 1 and 2). Over 90% of cadets "Agreed" or "Strongly Agreed" that the activities were useful to practice Teamwork (94.29%), Communication (94.29%), and Decision Making (91.43%). The majority of cadets also viewed the activities as useful to practice Flexibility (82.86%), Results Focus (71.43%), Accountability (62.86%), Developing Others (57.14%), and Self-Control (57.14%). Additionally, over 70% of cadets "Agreed" or "Strongly Agreed" that, based on the LLAB Minecraft activities, they received useful feedback on Teamwork (79.41%), Communication (77.14%), and Decision Making (71.43%). The majority of cadets also agreed that they received useful feedback on Results Focus (68.57%) and Flexibility (57.14%).

Notably, the activities were viewed as useful by cadets regardless of their level of prior experience with Minecraft or other online games. Among cadets who had never or rarely played Minecraft prior to participating in the DAGGER program, over 90% viewed the LLAB

Minecraft exercises as useful to practice Teamwork (100%), Communication (91.67%), Decision Making (91.67%), and Flexibility (91.67%), with the majority also viewing the exercises as useful to practice Results Focus, Accountability, and Self-Control (see Table 8). More than 70% of this group also indicated that they received useful feedback on Teamwork, Communication, Flexibility, and Results Focus based on the LLAB Minecraft activities. Overall the extent of cadets' prior Minecraft experience was not significantly correlated with either perceived usefulness to practice the Foundational Competencies (r = -.22) nor perceived usefulness for feedback on the Foundational Competencies (r = .09) (see Table9). Online gaming experience was generally unrelated to perceptions of the LLAB Minecraft activities (see Table9). However, those who more frequently used online building games outside of the DAGGER program were significantly less likely to view the LLAB Minecraft activities as useful for practicing the Foundational Competencies (r = -.41).

Narrative comments regarding the LLAB Minecraft exercises, sorted by theme, appear in Table 10. The majority of comments were positive, with cadets highlighting the LLAB Minecraft activities as useful to practice Foundational Competencies (N = 5), and indicating that use of the activities should be expanded (N = 3), and were engaging and immersive (N = 2). However, multiple cadets recommended more detailed instructions for how to execute the activities (i.e., more specific Rules of Engagement) (N = 4). A smaller number of negative comments indicated that the LLAB Minecraft activities were not sufficiently challenging or useful (N = 3) or that they encountered some technical issues with software or hardware that limited the number of cadets who could participate (N = 2). Although one cadet indicated that it was difficult to integrate cadets without prior Minecraft experience into the LLAB Minecraft activities, another cadet highlighted the integration of cadets without prior Minecraft experience as a positive

aspect of working together as a team ("It was really fun to be playing with very experienced Minecraft players and people with no experience. It was a lot of communication and helping to figure out how to explain concepts, Minecraft tools and controls so that no player was left behind.")

Cadet Survey: Perceptions of Peer Rating Tool. Many of the cadets who participated in the Minecraft team activities in LLAB had not used the web-based rating tool to provide feedback. Nevertheless, the majority of those who used the web-based rating tool indicated value in the rating process (see Table11). Over 60% of respondents indicated that the process of (i) self-rating and (ii) rating one's teammates in LLAB "helped me reflect on the Foundational Competencies" (66.66% and 63.16%, respectively). Additionally, overall 73.69% of respondents indicated agreement that "Having to provide comments to justify my ratings of teammates on each competency helped me reflect on the Foundational Competencies." Overall 63.16% of respondents indicated agreement that "Viewing how my teammates rated me on each competency in LLAB helped me better understand my competency strengths and weaknesses."

Notably, among those who participated in the LLAB Minecraft activities, cadets who used the web-based rating tool were significantly more likely than those who did not to indicate that they had received "useful feedback" on the Foundational Competencies based on the LLAB Minecraft exercises [$t(33) = 2.36$, $p < .05$]. Specifically, as shown in Table 12, cadets who used the web-based rating tool were significantly more likely to receive useful feedback based on the LLAB Minecraft activities on Communication and Flexibility [$t(33) = 3.05$, $p < .05$ and $t(33) = 2.28$, $p < .05$].

Discussion

In the absence of the physical space and resources required for many “hands-on” team activities, AFROTC detachments have had to think creatively about how they can provide opportunities for active learning that can help cadets improve in the areas of Teamwork, Communication, and other Foundational Competencies the Air Force recognizes as critical for success across an Air Force career. The two-year pilot DAGGER program considered whether online gaming (specifically, Minecraft Education, which AFROTC detachments already have access to through their universities) could provide a meaningful opportunity for cadets to work as a team and practice and receive peer feedback on the Foundational Competencies.

Results from the survey of cadets who participated in the program demonstrate that the use of Minecraft team activities (building, navigation, escape rooms) in AFROTC leadership courses was generally perceived positively. Cadets viewed these activities as useful to practice applying the Foundational Competencies (particularly, Teamwork, Communication, and Decision-Making), and as a useful as a basis for competency-based feedback. Cadets who used a standardized web-based competency rating tool in combination with the Minecraft activities were most likely to report receiving useful feedback, but other cadets also reported receiving useful competency feedback to a somewhat lesser extent.

The survey highlights that online team-based games, and Minecraft specifically, are widely used by AFROTC cadets, suggesting that such games may be motivating for many cadets if used in the context of military training and education. Specifically, over 75% of cadets surveyed reported having previously used Minecraft, and 80% reported having used some form of team-based online game. However, results also show that the extent of cadets’ experience with such games varies considerably, highlighting the importance of designing activities in a way that allows for the effective integration of cadets both with and without gaming experience. Our

results demonstrate that the Minecraft activities used in AFROTC Leadership Lab were generally viewed as useful for practicing and receiving feedback on competencies like Teamwork and Communication even among cadets who had never or only rarely used Minecraft or other online games.

Table 1

AFROTC Status of Student Survey Respondents

	All survey participants* (N = 149)	Participated in Minecraft Exercises in LLAB (N = 35)	Used Competency Rating Tool in LLAB (N = 19)
AS100 Cadet	35	1	2
AS200 Cadet	57	11	8
AS300 Cadet	24	11	1
AS400/700/800/900 Cadet	19	6	4
AS500 Cadet	10	4	1
Other DAF Member	4	2	3

Notes. Includes all respondents who completed the gaming experience survey; by policy, respondents who indicated they were not currently affiliated with AFROTC or DAF (e.g., students not currently enrolled in AFROTC) were ineligible and were immediately exited out of the survey. Other DAF member includes former AFROTC cadets who commissioned after participating in the DAGGER program.

Table 2.

Team Performance Comments from Higher and Lower Performing Teams, by Minecraft Activity Type (Det 150)

	High Performing Team (Team Performance Ratings of 4.75+)	Lower Performing Team (Team Performance Ratings <3.50)
Escape Room	• "[We] had strong **teamwork**, strong **leadership**, a **clear goal**, and the mission was accomplished within 12 minutes." • "**Role delegation** was effective and ensured that we were able to complete all the tasks efficiently. One **team goal** is to maintain the **clear communication** we had for this [exercise] in the next [exercise]." • "We got stuck with the sword puzzle at the end, but we **started thinking outside of the box** and accomplished the task" • "**[We thought] creatively and work[ed] together** as a team to accomplish the task at hand."	• "**We could have had better communication or critical thinking skills**, especially in the end when we took a while to figure out the code and the lever configuration. We took too much time focusing on the swords, which weren't relevant to the problem. **Once they failed** to work a few times, **we could have thought more quickly to go back** to the original levers like we did in the end." • "We completed the mission in the allotted time. There were no impactful issues that came up as we unlocked each room, however **each room took us a few tries to fully understand** how to progress. As a result, I believe our performance was fairly average." • "Team completed the mission though **could have done better working together as a team and thinking critically about the mission** at hand."
Navigation	• "[We] **successfully delegated tasks** while maintaining control over the scenario to ensure that we could meet all of the objectives in the time allotted. **Effective distribution of work** was very important here and should be utilized more often. We hope to finish in areas we did not complete and continue on through delegation to complete the next set of tasks." • "[We] successfully built the mission objectives. Different areas were **allocated to different people**. The mission was **accomplished within the time limit**. We learned that it was very effective to **distribute the work** required to finish the mission between the members of the team. One goal for the next activity is to continue to **effectively delegate roles** within the mission and to **continue to communicate effectively** in order to finish the mission." • "We successfully **build multiple structures and tasks to give teams opportunities to work together** to complete the mission together. Our goals would be to complete the areas, and continue to the next objectives."	• "The team successfully **started to create location objectives by having each member build a different part** of the "maze". We have more to add on in the next few sessions, but we successfully **laid a groundwork** to continue completing the objectives in the next two build sessions. Our goals for the next activity are to complete the entire maze, with the goal of refining it in the last session to ensure we meet all the objectives." • "Team had a **slow start** at the beginning as we were still **unsure of our objective** for this specific session and later on, what our design goals were. However, **afterwards** we were able to pick up the pace as we **solidified our plans and assigned roles**." • "The team **started off slow** in regards to working towards the objective, however were **ultimately able to create an initial plan of action** and move forward, **adapting and detailing the specifics of the plan as we worked through it** and experienced obstacles. Lesson learned is to **focus on just creating a general initial plan in situations that are more complex**. Goal for the next activity is to continue with the plans we created today and expand upon them towards the ultimate objective."

Table 3.

Example Comments Accompanying Peer Ratings of Teammates (Minecraft Navigation Activities) (Det 150)

	Competency	Top Rating (5)	Above Average Rating (4)	Average Rating (3)	Low Rating (2)
Navigation	Decision Making (1)	• "Able to **decide who is building what**, and the **starting and end points** of the building project"	• "**Aided in planning phase** and came up with second part of the objective."	• "**Made decisions on individual part** of the scenario and then progressed to help with other parts of the scenario. **Could have made more decisions in assigning roles for the team as well as plans for meeting our future objectives**."	• "**Took others opinions** and worked well within the time frame."
	Flexibility (2)	• "Was **able to adjust his design ideas** as required by the mission **in an efficient manner contributing in the successful completion** of the [activity]"	• "Made **timely decisions** and **effectively reacted to problems encountered** with the build."	• "**Adapted well to what was suggested by [another teammate]**."	• "**Was willing to change part** of the strategy **based on** time constraint and from **different opinions**."
	Teamwork (3)	• "Was **able to assign roles as leader** for team members and **fostered team work** during mission scenarios."	• "Was able to contribute to a large portion of mission completion while **aiding each of us in what we needed without taking over** what we were doing."	• "**Effectively communicated** with other team members and **collaborated on the build**. **Could have communicated more on the goals of the project** and our intentions with the build."	• "Did a good job accomplishing his specific role, working within the constraints, and **was a good follower** to the team lead."

Note. Ratings of 1 were extremely uncommon, and, upon review, typically did not address competency ratings based on the Minecraft activities themselves (e.g., "Forgot laptop for our meeting, resulting in us postponing [the activity] for an hour and a half.")

Table 4

Example Comments Accompanying Peer Ratings of Teammates (Minecraft Escape Room Activities) (Det 150)

	Competency	Top Rating (5)	Above Average Rating (4)	Average Rating (3)
Escape Room 1 and 2	Accountability (1)	• "**Took responsibility** for his leadership and **ensured that no one was left behind**."	• "**Recognized when we were lacking** in certain areas, when we were leaving someone, **and when we needed to get back on track**, took accountability as a follower as well."	• "Maintained general awareness of the location of the team, however **made no extra creative efforts to further accountability**."
	Communication (2)	• "**Effectively communicated the mission and instructions** for each section of the map." • "Great communication throughout the map, **giving ideas on how we could solve the problems**."	• "Communicated a **solution for one of the complex puzzles in a manner that left no room for confusion** or misunderstanding." • "**Communicated effectively to** accomplish the mission and to **determine where the boundaries are and how to complete/avoid each obstacle**."	• "**Sufficiently communicated personal steps taken** towards the goal and actions **with the leader before they were made**." • "Communicated during mission scenario, albeit not as often as others. **More communication would have been appreciated**."
	Self-Control (3)	• "**No signs of frustration or negative emotions** during execution"	• "**Maintained composure** and took the mission **seriously**."	• "Responded to changes in the scenario in a calm manner. **Did not overstep the leader or other teammates for the most part**. Began to overstep a teammate at one point due to a communication issue but fixed it after recognizing the issue."

Escape Room 3 and Cheyenne	Develops Others (1)	N/A	• "**Empowered fellow teammates to work together and come up with their own ideas** to solve problems presented during mission scenario."	• "**Contributed his knowledge** and gave specific instructions related to the mission on how to operate certain Minecraft features."
	Fosters Inclusion (2)	N/A	• "**Recognizes strengths and weaknesses** with the variety in the team **and used this to** build a better environment and **effectively accomplish tasks**."	• "Did a good job at including everyone else's ideas and encouraging everyone's roles. **Never excluded ideas**."
	Results Focus (3)	N/A	• "**Made sure to always be on task and worked efficiently throughout** scenario **to ensure that team met objectives**."	• "Had goal in sight, and **completed each goal with teammates."**

Table 5

Unique Words Across Self and Peer Rating Justifications, By Exercise and Competency

		Min and Max	Median	Mean	SD
Navigation	Decision Making (1)	9 to 83	40	38.76	21.17
	Flexibility (2)	12 to 79	34	36.64	17.88
	Teamwork (3)	7 to 78	31	35.16	21.11
Escape Room 1 and 2	Accountability (1)	16 to 62	27	33.44	15.20
	Communication (2)	13 to 61	39.5	37.94	15.40
	Self-Control (3)	7 to 51	25.5	29.63	13.02
Escape Room 3 and Cheyenne	Develops Others (1)	10 to 43	19	24.89	12.59
	Fosters Inclusion (2)	10 to 43	25	25.78	12.60
	Results Focus (3)	12 to 44	23	25.33	10.44

Note. The unique word count represents the total number of words that a rater used only once in justifying ratings of their self and each of three peers on the competency assessed within a given exercise. Using this metric, a rater who wrote the same comment for all three of his peers would have a lower unique word count than a rater who wrote comments that were more differentiated and personalized to individual peers. In practice, this metric also has the effect of excluding very common words such as “the,” “and,” “that,” etc. that often appear multiple times in a comment about a single individual.

Table 6

Prior Gaming Experience Among Current AFROTC Cadets (N = 145)

	Never	Rarely or Very Rarely	Occasionally	Frequently or Very Frequently
Minecraft	24.14%	16.55%	34.48%	24.83%
Online single-player games	17.93%	15.17%	24.14%	42.76%
Online team-based games	20%	14.48%	20%	45.52%
Online building games	23.61%	23.61%	24.31%	28.47%
Online escape rooms	44.14%	37.93%	10.34%	7.59%

Notes. N = 144 for the building game item. Item stem: "Prior to participating in the DAGGER program, how often had you used…"

Table 7

Cadet Perceptions of Competencies Developed Through Minecraft Team Activities (*N* = 35)

	Useful to Practice Competency		Received Useful Feedback on Competency	
	% Agreement	Mean (SD)	% Agreement	Mean (SD)
Accountability	62.86%	3.69 (1.02)	45.71%	3.46 (1.17)
Decision-Making	91.43%	4.14 (0.65)	71.43%	3.86 (0.88)
Developing Others	57.14%	3.54 (1.01)	48.57%	3.54 (0.98)
Communication	94.29%	4.51 (0.82)	77.14%	4.11 (0.90)
Flexibility	82.86%	4.03 (0.89)	57.14%	3.71 (0.99)
Self-Control	57.14%	3.71 (0.86)	48.57%	3.54 (1.04)
Teamwork	94.29%	4.40 (0.77)	79.41%	4.12 (0.88)
Results Focus	71.43%	3.97 (1.07)	68.57%	3.77 (0.84)

Notes. Participated indicated agreement or disagreement based on the following stems: "The Minecraft exercises I participated in during LLAB were useful team activities to practice…" and "Based on the Minecraft exercises I participated in during LLAB, I received useful feedback on…" Percentage agreement includes those who "Agreed" or "Strongly Agreed" with the statements. All usefulness ratings were scaled 1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree Nor Disagree, 4 = Agree, and 5 = Strongly Agreed.

Table 8

Perceptions of Competencies Developed Through Minecraft Team Activities, Among Cadets With Little or No Minecraft Experience (*N* = 12)

	Useful to Practice Competency		Received Useful Feedback on Competency	
	% Agree or Strongly Agree	Mean (SD)	% Agreement	Mean (SD)
Accountability	66.67%	3.69 (1.02)	33.33%	3.46 (1.17)
Decision-Making	91.67%	4.14 (0.65)	58.33%	3.86 (0.88)
Developing Others	41.67%	3.54 (1.01)	41.67%	3.54 (0.98)
Communication	91.67%	4.51 (0.82)	75%	4.11 (0.90)
Flexibility	91.67%	4.03 (0.89)	75%	3.71 (0.99)
Self-Control	58.33%	3.71 (0.86)	50%	3.54 (1.04)
Teamwork	100%	4.40 (0.77)	91.67%	4.12 (0.88)
Results Focus	75%	3.97 (1.07)	75%	3.77 (0.84)

Notes. Based on sub-sample who had “Never,” “Very Rarely” or “Rarely” played Minecraft prior to participating in the DAGGER program.

Table 9

Correlations Among Cadet Prior Gaming Experiences and Perceived Usefulness of LLAB Minecraft Activities (N = 34)

	1	2	3	4	5	6
1. Minecraft Experience	--					
2. Team-Based Online Game Experience	.68*	--				
3. Single-Player Online Game Experience	.68*	.91*	--			
4. Online Building Game Experience	.82*	.79*	.84*	--		
5. Online Escape Room Experience	.47*	.57*	.55*	.46*	--	
6. Perceived Usefulness for FC Practice	-.22	-.13	-.24	-.41*	.23	--
7. Perceived Usefulness for FC Feedback	.09	.13	.04	-.12	.12	.73*

Notes. *$p < .05$. Prior gaming experience variables were coded 0 = Never, 1 = Very Rarely, 2 = Rarely, 3 = Occasionally, 4 = Frequently, or 5 = Very Frequently. Perceived Usefulness of Minecraft Activities for Foundational Competencies Practice and Feedback were computed as the mean across the eight competency items (coded 1 to 5).

Table 10

Cadet Feedback Regarding LLAB Minecraft Activities, By Theme

Theme	# Cited Theme	Comments
Useful to Practice Foundational Competencies (Communication, Teamwork)	5	• "The Minecraft scenario was beneficial in strengthening teamwork and communication. The cool thing about it is your navigating different tasks in a virtual world while collaborating with your peers in real time, in the same room." • "I think the activities are a good exercise in teamwork, and I really enjoyed it." • "I think it was a great way for us to practice completing task separately and practice communicating with team members in detail. it also helped us practice quickly adapting to changes." • "It was really fun to be playing with very experienced Minecraft players and people with no experience. It was a lot of communication and helping to figure out how to explain concepts, Minecraft tools and controls so that no player was left behind." • "I think this is an innovative idea that can be useful in developing cadets' foundational competencies."
Provide More Detailed Instructions	4	• "I believe Minecraft-related GLPs could be more beneficial if they were to have a narrow focus and strict ROEs." • "I thought it was really good, but maybe before having to do this Minecraft led GLP, some rules of engagements or document explaining the game better would have been beneficial" • "If the Minecraft GLP were run better and we had a little more communication as to what it entails I think it would be great." • "I think it would've been beneficial to first have a crash course on Minecraft" • "Maybe have an explanation for what it really stands for so that we are all on same page"
Expand Use of the Activities	3	• "I appreciate how flexible GLPs developed by the DAGGER Program can be. I would like to see this program integrated into more training environments." • "I think increased amounts of activities might be beneficial to understanding and cooperation in group settings" • "I would like to see this tool utilized more"
Not Sufficiently Challenging or Useful	3	• "I believe that developing Airman and Guardians with the DAGGER program can be achieved. However, the program was not testing and challenging enough to be a useful technique of training." • "The challenges were too easy. Especially to people who were already familiar with Minecraft and video games as a whole." • "Wasn't very useful, just seemed like I was playing a video game"

Engaging and Immersive	2	• "Love the way we can immerse ourselves into the game more than regular GLPs." • "Fun and engaging, and even some of my cadre who had never played Minecraft were having a good time learning something new. One of my Cadre was very proud of his aquarium. :)
Technical Issues (Hardware/Software)	2	• "The issue I had with it was technical in nature. Our detachment has three different schools who participated in the activities. Since it is done through Minecraft Education, cadets across schools can't participate together…This seems like an unfortunate limiter and maybe would need coordination with Minecraft to resolve" • "I think the most challenging thing with these programs is having the hardware or the support to be able to implement these games. I had an okay time playing Minecraft on my current setup, but I know members of the program with i3 processors were struggling to run certain parts of the navigation course."
Difficult to Integrate Cadets Without Minecraft Experience	1	• "Problem with trying to get everyone to play. If they didn't play, hard for them to find a job to do."

Prompts: "Please provide any comments about your experiences with the Minecraft activities in LLAB" and "Please provide any additional comments about the Developing Airmen with Games for Enhanced Readiness (DAGGER) program" (comments about activities).

Table 11

Cadet Perceptions of LLAB Competency Rating Process

	"Strongly Disagree"	*"Disagree"*	*"Neither Agree Nor Disagree"*	*"Agree"*	*"Strongly Agree"*	Mean
"The process of rating myself in LLAB helped me reflect on the Foundational Competencies"	5.56%	11.11%	16.67%	33.33%	33.33%	3.78
"The process of rating my teammates in LLAB helped me reflect on the Foundational Competencies"	10.53%	10.53%	15.79%	36.84%	26.32%	3.58
"Having to provide comments to justify my ratings of teammates on each competency helped me reflect on the Foundational Competencies."	10.53%	5.26%	10.53%	47.37%	26.32%	3.74
"Viewing how my teammates rated me on each competency in LLAB helped me better understand my competency strengths and weaknesses."	5.26%	10.53%	21.05%	26.32%	36.84%	3.79

Notes. N = 19 for peer rating items, and N = 18 for the self-rating item.

Table 12

Cadet Perceptions of The Usefulness of Feedback Based on the Minecraft Exercises, Among Those Who Did and Did Not Use the Web-Based Rating Tool During LLAB

	"Based on the Minecraft exercises I participated in during LLAB, I received useful feedback on…"			
	Used Web-Based Rating Tool During LLAB (N = 8)		Did Not Use Web-Based Rating Tool During LLAB (N = 27)	
	% Agreement	Mean (SD)	% Agreement	Mean (SD)
Accountability	75%	4.00 (1.31)	37.04%	3.30 (1.10)
Decision-Making	100%	4.25 (0.46)	62.97%	3.74 (0.94)
Developing Others	75%	4.00 (1.07)	40.74%	3.41 (0.93)
Communication	100%	4.88 (0.35)	70.37%	3.89 (0.89)
Flexibility	75%	4.38 (0.92)	51.85%	3.52 (0.94)
Self-Control	75%	4.00 (1.07)	40.74%	3.41 (1.01)
Teamwork	100%	4.63 (0.52)	70.37%	3.96 (0.92)
Results Focus	100%	4.25 (0.58)	59.26%	3.63 (0.88)

"Based on the Minecraft exercises I participated in during LLAB, I received useful feedback on…"

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Appendix 1

Foundational Competency Definitions

1. Communication - Effectively presents, promotes, and prioritizes varies ideas and issues both verbally and non-verbally through active listening, clear messaging and by tailoring information to the appropriate audience.
2. Decision Making - Makes well-informed, effective, and timely decisions that weigh situational constraints, risks, and benefits.
3. Accountability - Demonstrates reliability and honesty; takes responsibility for the actions and behaviors of self and team.
4. Flexibility - The ability to adapt to and work with a variety of situations, individuals, or groups effectively.
5. Self-Control - Keeps emotions under control and restrains negative actions when under stress.
6. Develops People - Invests in others to maximize their contributions to the mission by inspiring and providing an environment of continual feedback and learning opportunities.
7. Teamwork - Collaborates effectively with others to achieve a common goal or complete a mission task.
8. Results Focused - Demonstrates concern for working well or for competing against a standard of excellence.

Appendix 2

Sample Foundational Competency Rating Criteria Presented to AFROTC Cadets (Adapted from Barron & Rolwes, 2020)

Competency: **Teamwork** Definition: Collaborates effectively with others to achieve a common goal or complete a mission task				
1	2	3	4	5
NOVICE: Does not yet meet criteria for Basic Level.	BASIC: Acknowledges contributions made by others on the team. Participates during team activities while working toward a goal.	INTERMEDIATE: Helps other team members work toward team goals. Freely shares information with others on the team.	ADVANCED: Acknowledges conflict and works to resolve issues. Ensures teams work together toward a common goal.	EXPERT: Models collaborative excellence and guides others to improve collaboration. Develops strategies to ensure team members remain focused on goals despite major obstacles. Anticipates conflict and works to resolve situations that could affect team goals.

Competency: **Results Focused** Definition: Demonstrates concern for working well or for competing against a standard of excellence				
1	2	3	4	5
NOVICE: Does not yet meet criteria for Basic Level.	BASIC: Ensures projects within areas of specific responsibility are completed in a timely manner. Meets established suspenses.	INTERMEDIATE: Accomplishes work projects diligently. Actively strives to make a positive contribution through one's efforts.	ADVANCED: Continually works towards a defined standard of excellence. Monitors and evaluates plans, focuses on results and measuring attainment of outcomes.	EXPERT: Sets challenging goals for team to increase organizational excellence. Formulates innovative strategies to achieve self-set goals and improve performance in all facets of work. Sets challenging goals to continually increase personal standards of excellence.

Appendix 3.

Example Minecraft Team Activities Used in AFROTC Leadership Lab

Sample Building Activity**:**

BUILDING REQUIREMENTS

Create a defensible base as a team. The size of the base is up to you to decide.

Choose a suitable location, and ensure that the following is enforced at a minimum:

- A barracks in which to quarter the base population
- An armory to store your weapons and ammunition
- A mess hall/dining facility in which to store/distribute food
- Security perimeter around the entire base
- At least two entry/exit points, located in different sections of the base
- Any additional fortifications as deemed necessary to provide defenses, such as observation towers, barricades, etc.
- A library building that contains "Competency Message Walls," which are individual walls with messages containing tips about how to better develop each competency. EVERY TEAM MEMBER should contribute at least one message per wall using the sign block inside Minecraft.

Sample Escape Room Activity:

DESCRIPTION

In this Minecraft activity, each team will work together to complete the escape room before the timer runs out.

You are a Paratrooper tasked with saving an ally behind enemy lines. Hop on board the cargo plane and begin your mission to work through the jungle undetected together as a team.

INSTRUCTIONS

Have one officer trainee/cadet per team act as the Minecraft host, who should import the world map (if needed), start a multiplayer game as a host, and share the share code with all teammates.

Go through the escape room with your team ◦

Have one person set a timer to 20 minutes (or other duration specified by your instructor). Stop when the timer ends.

MAP LINK

Use the Escape Room 1 (4 player) map if your team has 4 total players. Use the Escape Room 1 (3 player) map if your team has 3 total players.**Appendix 4.**

Recommended Session Workflow.

1. Session Information: Make sure each member of your team knows which specific Session you will be doing.
2. Activity: All team members should be prepared for the activity, including all necessary software if needed (e.g., video game software for a gaming activity).
3. Competencies: Make sure that all team members know the competencies to focus on during the activity. Ideally, have a short discussion about each competency, e.g., share real-life examples with each other.
4. Conduct Activity: Complete your team activity for the specific Session. The activity should ideally take around 20-30 minutes. Longer activities can be divided into multiple Sessions.
5. Calibration Evaluation: Open the app and start a new Calibration
    a. Complete each tab by filling in required information.
    b. On the Rate Skills tab, make sure to seriously reflect on your and your teammates competency levels.
    c. (IMPORTANT!) Please make sure that the comments you write aligns with the competency rating that you selected
6. Reviewing Dashboard : After submitting each Calibration Evaluation, review your updated personal dashboard that displays specific metrics about your foundational competencies.
    a. Justification Score (JS): JS refers to how well a user can justify a specific skill rating by providing valid and meaningful comments that support that skill rating. JS is a measure of skill level awareness.
    b. Calibration Score (CS): CS is a measure of skill consensus between self and peers. CS is a measure of team norming and skill rating accuracy.
    c. Mastery Score (MS): MS is a measure of a user's overall skill level of the specific skill in the current context. MS is a measure of actual skill growth in a specific context.
7. Reflect upon your dashboard metrics, and prior to the next Session, discuss with teammates about competencies that you or your team needs to focus on more.

Figure 1.

Trainees Using Minecraft Within Classroom (Separate Devices, In-Person Interaction)

Figure 2.

Example Screenshot: Providing Peer Feedback in the Web-Based Rating Tool

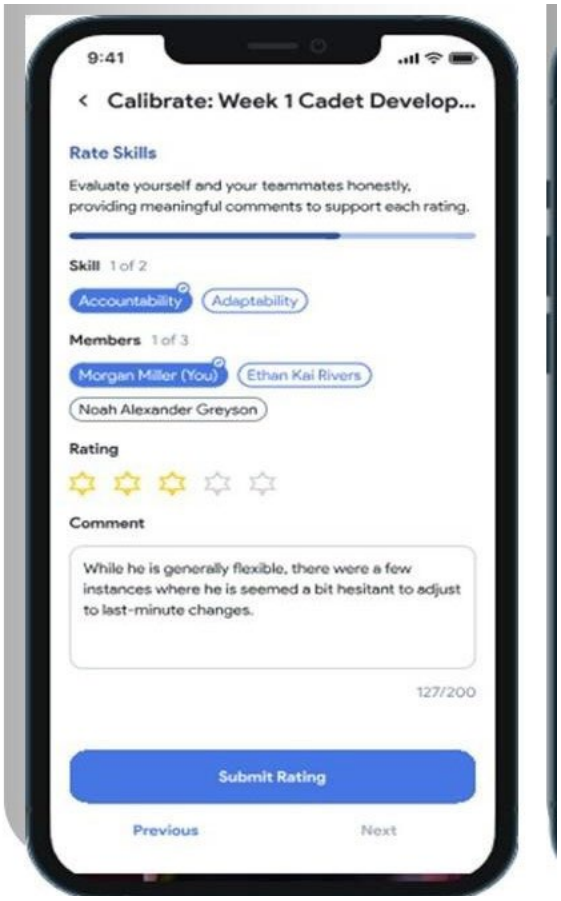
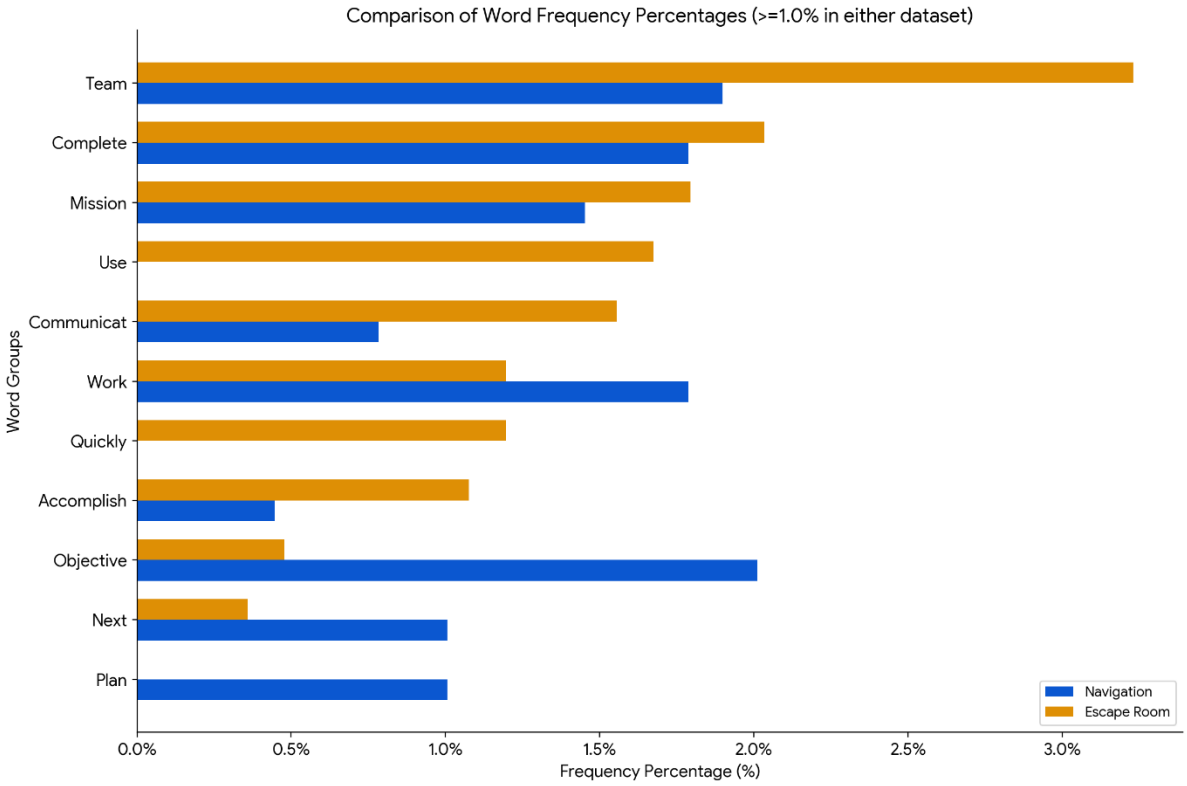


Figure 3.

Word Frequency in Cadet Descriptions of Team Performance in the LLAB Minecraft Activities.



Note. To provide a clearer analysis, we removed articles (e.g., “the,” “a”) and pronouns (e.g., “I,” “we”) and combined directly related words into groups. Groups with more than one word variation were: Accomplish (accomplish, accomplished), Build (build, built, building), Communicat (communication, communicated, communicating, communicate), Complete (complete, completed, completion, completing), Continu (continue, continued), Creat (create, created, creative, creating, creatively), Decod (decode, decoded, decoding), Delegat (delegate, delegated, delegation), Effectiv (effective, effectively, effectively), Efficien (efficient, efficiently), Figur (figure, figured, figuring), Finish (finish, finished), Full (full, fully), Goal (goal, goals), Learn (learned, learn), Objective (objective, objectives), Plan (plan, plans, planning), Progress (progress, progressed, progressing), Puzzl (puzzle, puzzles), Result (result, results), Session (session, sessions), Story (story, storyline), Task (task, tasks), Team (team, teams, teamwork, teammates), Think (think, thinking, thought), Use (use, using, used), and Work (work, worked, working). Word groups that comprised at least 1.0% of the words for either activity type are shown (836 words in the original unfiltered text for escape room activities and 895 for navigation activity text).